NGAME Terminology Glossary

Four-Layer Model for Churn, Detection, Interpretation, and Adjudication

Status: Draft for journal manuscript
Purpose: Assign precise, layer-specific vocabulary across the NGAME pipeline; retire confusion-matrix labels (TP / FP / FN / TN) on the Jaccard 2×2 (Layer 1); separate statistical detection (Layer 2), LLM taxonomy linkage (Layer 3), and FRP adjudication (Layer 4).
Audience: Co-authors, reviewers, and readers of the NGAME design-science evaluation.

Why This Glossary Exists

NGAME uses “positive,” “negative,” “reliable,” and “triage” in several distinct senses. Conflating them invites reviewer criticism and misstates what the artifact does. This glossary assigns **one vocabulary per layer** and directs each layer’s **trust property** — the kind of warrant the layer can honestly claim — without imposing a single global definition of “reliability.”

For the anticipated reviewer question **“What about false positives and false negatives?”** see **Tables A–E** (below the layer overview).

Layer	Retires or restricts
1 — Structural	TP/FP/FN/TN on the day-over-day Jaccard 2×2; use retained / added / removed / excluded
2 — Detection	Standard IR / confusion-matrix terms **only** in controlled evaluation against ground truth
3 — Scheme interpretation	“Triage,” “reliable ranking,” and fraud-verdict language for LLM taxonomy output
4 — FRP adjudication	Unqualified “true/false positive” and autonomous fraud verdicts

Layer Overview

Layer	Question answered	Primary outputs	Ground truth	Trust property
1 — Structural	How stable is each transaction-type population from yesterday to today?	CPI, α (addition rate), δ (deletion rate), wCPI	None required; descriptive set overlap	**Reproducibility** — same snapshots → same CPI
2 — Detection	Which φ dimensions deviate enough from the trained baseline to rank in the top-*k* alert set?	z-scores, HIGH / MEDIUM / LOW bands, top-*k* rankings	Injected or known anomalies (evaluation only)	**Detection validity** — precision/recall on dimension ranking (eval. only)
3 — Scheme interpretation	Given flagged dimensions, how plausibly do patterns align with asset-misappropriation scheme classes?	Taxonomy mappings, LLM narrative, model-reported confidence, deterministic anchor scores	Curated ontology + ACFE priors; not adjudicated fraud	**Interpretive fidelity** — grounded, auditable, non-authoritative
4 — FRP adjudication	What did human review conclude, and what action followed?	Corroborated / benign / missed outcomes; management warnings as briefing artifacts	FRP professional judgment after review	**Adjudication outcome** — operational ground truth in production

Pipeline order: Layer 1 → Layer 2 → Layer 3 → Layer 4. Layer 2 produces the **statistical alert set** (deterministic top-*k* ranking). Layer 3 **enriches** that set with scheme labels and narrative; it does not replace Layer 2 as the primary trigger for FRP attention. Layer 4 is where "what actually happened" is established.

Answering "But is it reliable?"

Readers will ask **"Is it reliable?"** at every layer. Do not answer with one global yes or no. Point to the **trust property** for that layer:

Layer	Reader's question	Answer in one line
1	Is CPI measurement trustworthy?	Same Y/T inputs yield the same CPI; descriptive, not evaluative.
2	Does it flag the right ϕ dimensions?	Top-*k* ranking is deterministic given μ, σ; precision/recall apply under controlled evaluation.
3	Is the scheme label / LLM narrative trustworthy?	Grounded in `Asset_Misappropriation.ttl`, auditable against a deterministic anchor; **not** a fraud verdict and not evaluated with Layer 2 IR metrics.
4	Was the management action correct?	FRP judgment is ground truth in production; outcomes are corroborated, benign, or missed.

Answering "What about false positives and false negatives?"

Reviewers and readers trained in machine learning will ask this question early. **NGAME does not retire confusion-matrix language globally.** It **relocates** TP, FP, FN, and TN to the layer where ground truth and a defined detection task exist, and **replaces** those labels elsewhere with terms that do not imply classifier error.

Direct answer (one paragraph for manuscript adaptation):

> False positives and false negatives are standard information-retrieval terms for comparing a detector's output to **known ground truth**. In NGAME they apply **only at Layer 2**, where top-*k* transaction-type dimensions are evaluated against injected or known anomalies. They do **not** apply to the Layer 1 Jaccard partition (retained, added, removed), which is descriptive set overlap without a reference "correct" class. At Layer 3, scheme labels are hypotheses auditable against a curated taxonomy anchor—not TP/FP outcomes. At Layer 4, an alert that proves benign on FRP review may resemble a Layer 2 false positive under a

> fraud-only ground-truth definition, but under NGAME's triage design it is a **benign resolution**, not evidence of broken detection.

Table A — Where TP / FP / FN / TN belong (and where they do not)

Layer	Do TP / FP / FN / TN apply?	Ground truth required?	Correct vocabulary
1 — Structural	**No**	No	Retained (*a*), Removed (*b*), Added (*c*), Excluded
2 — Detection	**Yes** (evaluation only)	Yes — injected or known anomalies	TP, FP, FN, TN (eval.); precision, recall, FPR
3 — Scheme interpretation	**No**	Partial — ontology + ACFE priors, not adjudicated fraud	Scheme-concordant mapping, interpretive divergence, narrative adjunct
4 — FRP adjudication	**No** (avoid unqualified use)	FRP judgment in production	Corroborated alert, benign resolution, missed event

Table B — What the reader may call a "false positive" vs what NGAME means

Reader's informal claim	Layer	NGAME term	Why the distinction matters
"A new vendor is a false positive on the Jaccard grid."	**1**	**Added** (*c*)	Layer 1 does not judge correctness; new records are structural facts, not errors.
"CPI flagged Vendors but no fraud was injected."	**2**	**FP (eval.)** — if and only if ground truth is defined	Valid eval. metric under controlled test; requires explicit anomaly injection record.
"CPI flagged Vendors but FRP	**4**	**Benign resolution**	Statistically warranted alert + legitimate business

Reader's informal claim	Layer	NGAME term	Why the distinction matters
found legitimate cleanup."			explanation; **not** a detector fault under triage design.
"The LLM named the wrong fraud scheme."	**3**	**Interpretive divergence**	Scheme label is a hypothesis, not a detection verdict; audit against deterministic anchor.
"The LLM said HIGH risk but there was no fraud."	**3 + 4**	**Benign resolution** (after FRP)	LLM enriches briefing; Layer 2 triggered review; FRP adjudicates.
"Fraud occurred but NGAME stayed quiet."	**2 + 4**	**FN (eval.)** if ground truth known; **missed event** in production	FN requires eval. ground truth; missed event is the operational Layer 4 label.

Table C — Layer 1 retired labels (Jaccard 2×2): why "false" is misleading

Training materials sometimes overlay confusion-matrix names on the Jaccard grid. **Do not carry that overlay into manuscript prose.**

Retired label	Was used to mean	Problem if kept	Use instead
TP	Retained	Implies a "correct detection" where no detector exists	**Retained** (*a*)
FP	Added	Implies a **new record is an error**; legitimate vendors become "false alarms"	**Added** (*c*)
FN	Removed	Implies a **missed detection** where churn is a structural fact	**Removed** (*b*)
TN	Excluded	Implies correct rejection in a classifier sense	**Excluded** (omitted from Jaccard denominator)

Same grid, different semantics: The 2×2 layout matches intersection-over-union (IoU) in segmentation, where TP/FP/FN are defined against a **reference mask**. NGAME Layer 1 compares **yesterday vs today** with no reference mask and no prediction task. CPI = a/n is Jaccard set similarity, not classifier accuracy.

Table D — Layer 2 detection task (the only home for standard FP / FN metrics)

Detection task (evaluation): Rank each of *N* ϕ dimensions by |z| (or composite alarm); treat the top-*k* as the alert set (*k* = 3; *N* = 17–18).

Term	Definition
TP (eval.)	Flagged dimension where an anomaly was knowingly introduced or present in ground truth
FP (eval.)	Flagged dimension with only normal variation relative to ground truth
FN (eval.)	Ground-truth anomaly dimension **not** in the top-*k* alert set
TN (eval.)	Normal-variation dimension not flagged
Precision	TP / (TP + FP)
Recall (top-*k*)	TP / (TP + FN)
False positive rate	FP / (FP + TN)

Scope: Metrics measure **transaction-type dimension ranking**, not dollar amount of a single transaction and not LLM scheme labels. Pipeline pass/fail alone is insufficient for formal precision/recall; per-pattern top-*k* logging against ground truth is required.

Table E — Layer 4 outcomes that resemble—but are not—the same as—Layer 2 FP / FN

Layer 4 outcome	Resembles (informally)	Is it the same as Layer 2 FP or FN?
Benign resolution	“False alarm” / FP	**No** — alert was statistically warranted; FRP established legitimate context. Often **would** count as FP (eval.) only if ground truth were strictly “no anomaly of any kind.”
Corroborated alert	“True hit” / TP	**Aligns with TP (eval.)** when eval. ground truth exists; broader in production (includes control weaknesses, not only fraud).
Missed event	“Miss” / FN	**Aligns with FN (eval.)** when ground truth is known; operational label when fraud was material

Layer 4 outcome	Resembles (informally)	Is it the same as Layer 2 FP or FN?
		but did not surface in top-*k* or warnings.

Design intent: NGAME is a **triage instrument**. A moderate Layer 2 FP rate under fraud-only ground truth is acceptable when the cost of late detection (ACFE-style median lag without continuous monitoring) dominates the cost of FRP review of statistically warranted alerts.

Suggested manuscript paste-in (false positives / negatives)

> *False positives and false negatives:* Confusion-matrix terms apply only to Layer 2 evaluation of top-*k* ϕ-dimension ranking against known or injected anomalies. Day-over-day record overlap (Layer 1) uses retained, added, and removed—not TP, FP, FN, or TN—because it is descriptive Jaccard similarity, not classifier output. Scheme interpretation (Layer 3) produces taxonomy-grounded hypotheses, not FP/FN outcomes. At Layer 4, alerts resolved as benign after FRP review are not classified as false positives in the detector-error sense; they are **benign resolutions** of statistically warranted signals.

Table 1 — Layer 1: Day-over-Day Record Overlap (Jaccard / CPI)

Replaces the Jaccard 2×2 labels TP, FP, FN, TN in manuscript prose.

Rows: presence in **Yesterday's** snapshot (Y). Columns: presence in **Today's** snapshot (T).

Cell	Preferred term	Symbol	Set notation	Role in CPI	Interpretation
Upper-left	**Retained**	*a*	$Y \cap T$	Numerator of CPI	Record URI present in both snapshots; continuity across days
Upper-right	**Removed**	*b*	$Y \setminus T$	Contributes to δ (deletion rate)	Record existed yesterday, absent today; churn or deletion
Lower-left	**Added**	*c*	$T \setminus Y$	Contributes to α (addition	Record absent yesterday, present

Cell	Preferred term	Symbol	Set notation	Role in CPI	Interpretation
				rate)	today; new entry
Lower-right	**Excluded**	—	(not counted)	**Not in formula**	Absent both days; universe of non-occurring records is undefined

Core formulas (Layer 1 only):

Quantity	Formula	Range
CPI (Churn Persistence Index)	CPI = a / n, where $n = a + b + c$	[0, 1]
Addition rate (α)	$\alpha = c / n$	[0, 1]
Deletion rate (δ)	$\delta = b / n$	[0, 1]
Partition identity	CPI + α + δ = 1	—

Edge convention: When $n = 0$ (no records of type k in either snapshot), CPI = 1.0 — absence is treated as stable.

Dollar-weighted variant: Same structure with dollar weights w_i instead of URI counts → wCPI, wAdditionRate, wDeletionRate.

Jaccard vs confusion-matrix layout (reader bridge)

In mainstream usage, Jaccard similarity is a **set-similarity coefficient**: |A ∩ B| / |A ∪ B|. The 2×2 partition (present/absent in Y × present/absent in T) has the **same grid shape** as a confusion matrix and as intersection-over-union (IoU) in segmentation, where cells are sometimes labeled TP/FP/FN relative to a **reference mask**. NGAME Layer 1 uses **no such reference**: rows and columns index **temporal membership** (yesterday vs today), not predicted vs correct. CPI = a/n is descriptive churn measurement, not classifier performance.

Retired Layer 1 labels (do not use in journal prose)

Retired label	Was used to mean	Use instead
TP (True Positive)	Retained	**Retained** (*a*)

Retired label	Was used to mean	Use instead
FP (False Positive)	Added	**Added** (*c*)
FN (False Negative)	Removed	**Removed** (*b*)
TN (True Negative)	Excluded	**Excluded** (state explicitly that this cell is omitted from Jaccard)

Note: “Added” is not a “false” outcome. A legitimate new vendor is **Added** at Layer 1 and may still yield a LOW z-score at Layer 2 if consistent with the trained baseline.

Table 2 — Layer 2: Detector Evaluation Metrics (Ground-Truth Comparison)

Use standard IR / confusion-matrix terms only here, when comparing NGAME’s top-*k* flagged ϕ dimensions to known or injected anomalies.

Detection task (evaluation): Given an *N*-element daily ϕ (CPI) array, rank dimensions by |z| (or composite alarm) and treat the top-*k* as the alert set (*k* = 3 in current instantiation; *N* = 17–18).

Term	Symbol	Definition	Formula
True Positive (eval.)	TP	Flagged dimension where an anomaly was knowingly introduced or present in ground truth	—
False Positive (eval.)	FP	Flagged dimension with only normal variation relative to ground truth	—
False Negative (eval.)	FN	Ground-truth anomaly dimension not in the top-*k* alert set	—
True Negative (eval.)	TN	Normal-variation dimension not flagged	—

Term	Symbol	Definition	Formula
Precision	—	Share of flagged dimensions that match ground truth	TP / (TP + FP)
Recall (top-*k*)	—	Share of anomalous dimensions captured in top-*k*	TP / (TP + FN)
False Positive Rate	FPR	Share of normal dimensions incorrectly flagged	FP / (FP + TN)

Scope limits (state in manuscript):

- Metrics apply to **transaction-type dimension ranking**, not dollar amount of any single transaction.
- Formal element-level precision/recall requires per-pattern logging of top-*k* rankings against ground truth; pipeline pass/fail alone is insufficient.
- Design intent: the alert instrument favors **recall over precision** given ACFE-style detection lag without continuous monitoring.

Layer 2 outputs (operational, pre-interpretation):

Term	Definition
z-score	$z_k = (CPI_k(\text{today}) - \mu_k) / \sigma_k$; signed deviation from trained mean
Deviation band	$\|z\| > 3.0$ → HIGH; $\|z\| > 2.0$ → MEDIUM; else LOW
Composite alarm	Fires when count-based or dollar-weighted signal reaches MEDIUM or HIGH
Top-*k* alert set	The *k* ϕ dimensions with largest $\|z\|$ (or equivalent ranking rule) on a given day; **primary statistical trigger** for downstream steps

Table 3 — Layer 3: Scheme Interpretation (Taxonomy Linkage)

Question: Given the Layer 2 alert set, how plausibly do observed patterns align with known asset-misappropriation scheme classes?

After top-*k* dimensions are flagged, NGAME maps associated accounts to classes in `Asset_Misappropriation.ttl` using a locally hosted LLM with retrieval-style prompting

(`ngame_llm_analysis_agent.py`). A **deterministic anchor** — curated QuickBooks transaction-type → taxonomy mapping with ACFE evidence priors (`ngame_misappropriation_mapping_and_scoring.py`) — provides a reproducible baseline for audit.

NGAME does not treat Layer 3 output as a fraud verdict. Scheme labels and narrative **brief the FRP**; they do not alone escalate severity or replace Layer 2 alerting.

Term	Definition
Statistical alert set	Top-*k* ϕ dimensions from Layer 2; primary trigger for FRP attention
Scheme-concordant mapping	Taxonomy class assignment consistent with both the flagged transaction type and curated ACFE/QBO priors
Taxonomy grounding	Assigned scheme classes must exist in `Asset_Misappropriation.ttl` ; no free-form scheme invention
Deterministic anchor	Reproducible misappropriation-likelihood score from curated mapping rules and ACFE evidence weights
Interpretive divergence	LLM-assigned scheme class or emphasis disagrees with the deterministic anchor
Narrative adjunct	LLM explanation text supporting FRP review; subordinate to statistical alerting
Model-reported confidence	LLM self-assessed confidence (0–1 in prompts); **epistemic, not calibrated** — not treated as probability of fraud

Warranted claims at Layer 3 (interpretive fidelity)

State explicitly what Layer 3 **does** warrant:

1. **Taxonomy grounding** — outputs reference ontology-defined scheme classes.
2. **Reproducibility under fixed model and temperature** — repeated runs with identical inputs are expected to be substantially stable, not necessarily bit-identical.
3. **Concordance with deterministic anchor** — LLM outputs can be audited against the curated mapping/scoring module.
4. **Non-authoritative role** — Layer 2 triggers review; Layer 3 enriches the briefing.

Layer 3 evaluation (lightweight; not a confusion matrix)

If quantitative reporting is required without a fourth metric grid:

Check	What it measures
Grounding rate	Fraction of LLM-assigned taxonomy classes that exist in the TTL
Anchor concordance	Fraction of flagged transaction types where LLM top scheme class matches deterministic anchor top class
Rerun stability	Qualitative agreement of scheme classes across repeated runs (fixed model, temperature)

Formal calibration of scheme labels against adjudicated outcomes (Layer 4 → retrospective labels) may be deferred; state as a limitation if not yet measured.

Terms to avoid at Layer 3

Avoid	Reason	Prefer
The triage (for LLM output)	Collides with Layer 4 FRP adjudication	**Scheme interpretation** or **taxonomy enrichment**
Reliable ranking (LLM)	Implies Layer 2–style detection validity	**Concordant with deterministic anchor** or **grounded scheme mapping**
False positive (scheme label)	Conflates with Layer 2 eval.	**Interpretive divergence** or **scheme mismatch on review**
Fraud confirmed	NGAME does not adjudicate	**Scheme-concordant hypothesis** (FRP decides at Layer 4)
Reliable (unqualified, for LLM)	Reader imports Layer 1–2 meaning	**Grounded**, **auditable**, or **interpretively concordant**

Table 4 — Layer 4: FRP Adjudication Outcomes (Operational)

NGAME does not accuse anyone of fraud. A management warning means activity is statistically unusual relative to the organization’s 30-day baseline; the FRP adjudicates at Layer 4.

Term	Definition	Relation to prior layers
Statistically warranted alert	$\lvert z \rvert$ or composite alarm exceeds threshold; signal is real relative to trained μ, σ	Layer 2; may correspond to FP (eval.) if ground truth is "normal business change"
Scheme-concordant hypothesis	Layer 3 mapping plausibly names an ACFE-style scheme class for flagged activity	Layer 3; may still yield **benign resolution** at Layer 4
Corroborated alert	FRP review finds material concern: fraud, control weakness, or required correction	Aligns with TP (eval.) when ground truth exists; broader in production
Benign resolution	Alert correctly reflected baseline deviation; FRP explains as legitimate (e.g., new GL accounts, seasonal vendor onboarding)	Often FP (eval.) under fraud ground truth; **not** a system failure under triage design
Missed event	Material misappropriation occurred but did not surface in warnings or top-*k*	FN (eval.) when ground truth is known

Terms to avoid at Layer 4

Avoid	Reason	Prefer
False positive (unqualified)	Implies detector error; conflates with Layer 2	**Benign resolution** or **statistically warranted, benign on review**
True positive (unqualified)	Implies fraud confirmed by the system	**Corroborated alert** (after FRP review)
Unreliable warning	Suggests broken instrumentation	**Benign resolution** or **spurious for fraud purposes, valid for triage**
Reliable (for warnings)	Collides with "reliable μ and σ estimates" (Layer 1 training)	**Corroborated**, **warranted**, or **actionable after review**

Table 5 — Cross-Layer Mapping (Illustrative)

One day's Vendors dimension can illustrate how layers diverge:

Layer	Example reading
1 — Structural	High **Added** and **Removed** counts → low CPI (churn)
2 — Detection	$\|z\| > 3$ on Vendors → HIGH; Vendors in top-*k*
3 — Scheme interpretation	LLM maps to fictitious-vendor / billing-scheme class; deterministic anchor agrees Vendors is high-evidence; narrative cites vendor churn
4 — FRP adjudication	FRP reviews Audit Log → **benign resolution** (legitimate vendor cleanup) or **corroborated alert** (shell-vendor pattern)

Chart of Accounts example (field deployment): Layer 2 correctly flags HIGH deviation; Layer 3 may propose a master-data scheme hypothesis; Layer 4 **benign resolution** after FRP confirms legitimate new GL accounts. Not an eval. FN; adjudication behaved as designed.

Scheme-concordant but benign: Layer 3 can be scheme-concordant while Layer 4 is benign — the statistical signal was real and the scheme hypothesis was plausible, but FRP established legitimate business context. That is not interpretive or detection failure.

Table 6 — Notation Quick Reference

Symbol	Meaning
Y, T	Yesterday and Today ontology snapshots for a transaction type
a, b, c, n	Retained, removed, added, and union count ($n = a + b + c$)
ϕ_k, CPI_k	Churn Persistence Index for transaction type k
μ_k, σ_k	Trained mean and standard deviation of CPI_k (30-day baseline)
$wCPI_k$	Dollar-weighted CPI for type k
$SCPI_k(t, L)$	Sequence CPI: geometric mean of CPI_k over window L
z_k	Standardized deviation: $(CPI_k(\text{today}) - \mu_k) / \sigma_k$
k (alert)	Number of top-ranked anomalous dimensions in evaluation (default 3)

Symbol	Meaning
FRP	Financially Responsible Person — designated reviewer

Suggested Manuscript Footnote (optional paste-in)

> *Terminology:* Day-over-day record overlap (retained, added, removed) describes structural churn only and is not labeled with confusion-matrix terms. True and false positives, precision, and recall apply solely to evaluation of top-*k* dimension ranking against known or injected anomalies (Layer 2). Scheme interpretation (Layer 3) maps flagged activity to ontology-defined misappropriation classes via a locally hosted LLM and is auditable against a deterministic curated anchor; it does not render fraud verdicts. The Financially Responsible Person adjudicates outcomes at Layer 4 as corroborated, benign on review, or missed.

Draft aligned with NGAME CPI analysis (`ngame_cpi_analysis_agent.py`), churn comparison thresholds (`ngame_churn_comparison_agent.py`), LLM taxonomy linkage (`ngame_llm_analysis_agent.py`), deterministic scheme scoring (`ngame_misappropriation_mapping_and_scoring.py`), FRP Operations Guide, and DSRM Evaluation Draft §5.2.
